



Fundamental electro-optic limitations of thin-film lithium niobate microring modulators

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Abstract: We investigate the impact of waveguide curvature on the electro-optic efficiency of microring resonators in thin-film X-cut or Y-cut lithium niobate (in-plane extraordinary axis) and derive explicit relations on the response. It is shown that such microring modulators have a fundamental upper bound on their electro-optic performance ($\sim 50\%$ filling factor) which corresponds to a specific arrangement of metal electrodes surrounding the microring and yields nearly identical results for X-cut and Y-cut designs. We further show that this limitation does not exist (i.e., 100% filling factor is possible) with Z-cut microring modulators or can be circumvented (i.e., $\sim 100\%$ filling factor is possible) in X-cut and Y-cut modulators that use a race-track configuration with segmented electrodes. Comparison of our analytical results with multiphysics simulations and measured electro-optic efficiencies of microring resonators in the literature demonstrates the validity and accuracy of our approach.

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1. Introduction

Thin-film lithium niobate on insulator (LNOI) has recently emerged as a promising platform for developing high-performance integrated photonic devices [1,2] due to the excellent electro-optical and nonlinear [3,4] properties of lithium niobate (LN) [5,6]. Examples include hybrid Si-LN waveguides [7], hybrid SiN-LN waveguides [8], heterogeneous chalcogenide-LN devices [9], and partially etched homogeneous LN devices [10,11].

The Pockels nonlinear effect in LN exploited with the use of lateral electrodes on both sides of a waveguide leads to the linear electro-optic response of LN-based phase shifters [8,10,12–16]. Due to its ease of fabrication, such lateral arrangement of electrodes is desirable and conveniently used with X-cut [10] or Y-cut LN [15] where the extraordinary optical axis is in plane and alignable to the applied lateral DC/RF field for maximizing the interaction with the optical mode (utilizing electro-optic $r_{33} \approx 30$ pm/V coefficient) inside the LN-based waveguide. Despite the pragmatism of this strategy, the electro-optic response deteriorates if the applied electric field is not aligned with the extraordinary axis of the crystal. This is unavoidable in microring-based structures [4,10,17], where optical waveguides bend across a range of in-plane orientations and the electro-optic response can significantly degrade depending on the range. Figure 1 illustrates the case in which a straight waveguide and its side electrodes are rotated by an angle ϕ . In this case, the applied lateral DC field still strongly interacts with the transverse electric (TE) mode of the waveguide (considering a rectangular cross section for the waveguide) but the r_{33} coefficient cannot be fully harnessed to produce maximum net electro-optic effect.

In this work, we first rigorously derive the impact of orientation on the LN-based phase shifters and then quantify the impact of curvature of microring/race-track resonators on their electro-optic

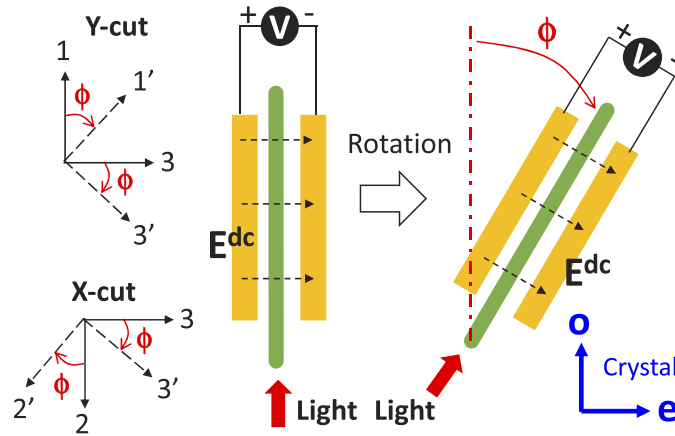


Fig. 1. Schematic of a straight LN waveguide with two electrodes on the side. The applied DC field between the two electrodes is exactly in the extraordinary direction of the crystal. Under the rotation, the applied DC field is no longer in the extraordinary direction. The 1, 2, 3 axes are aligned with the LN crystal coordinate system such that axis 3 is the extraordinary direction (optic axis). The axes under rotation by ϕ are denoted by 1', 2', 3'.

tuning with a coefficient called the *filling factor*. We show that the upper bound of the filling factor for a single-pair electrode arrangement in a microring modulator is ~ 0.25 which is achieved when the electrodes are symmetrically extended by 90° about the extraordinary direction. We also demonstrate good agreement between our analytic approach and the multiphysics simulations, as well as reported measured electro-optic efficiencies for LN-based microring modulators.

2. Electro-optic efficiency under rotation

In order to quantify the impact of the LN waveguide orientation on the electro-optic effect, we use the following notation in Fig. 1: the axes are labeled 1, 2, 3 such that the in-plane axes are 1 and 3 (Y-cut) or 2 and 3 (X-cut) where the extraordinary axis is 3. The unit vectors of the coordinate system are denoted by $\hat{1}$, $\hat{2}$ and $\hat{3}$. Under the rotation about the 2-axis for the Y-cut, the coordinate system is transformed into 1', 2', 3' such that $\hat{2} = \hat{2}'$ and the following relations hold:

$$\begin{aligned} \hat{1} \cdot \hat{1}' &= \cos \phi, & \hat{1} \cdot \hat{2}' &= 0, & \hat{1} \cdot \hat{3}' &= -\sin \phi \\ \hat{2} \cdot \hat{1}' &= 0, & \hat{2} \cdot \hat{2}' &= 1, & \hat{2} \cdot \hat{3}' &= 0 \\ \hat{3} \cdot \hat{1}' &= \sin \phi, & \hat{3} \cdot \hat{2}' &= 0, & \hat{3} \cdot \hat{3}' &= \cos \phi \end{aligned} \quad (1)$$

Now assume that the applied DC field is exactly aligned with the 3-axis. The change in the effective index of the optical mode of the waveguide before the rotation (i.e., $\phi = 0$) can be approximated by utilizing the perturbation theory [18]:

$$\Delta n_{\text{eff}}^{(0)} \approx \frac{c\epsilon_0}{4P} \iint_{\text{core}} \vec{\mathbf{E}}_{\text{opt}} \cdot \Delta \epsilon \vec{\mathbf{E}}_{\text{opt}}^* dS, \quad (2)$$

where $\epsilon_0 = 8.85 \times 10^{-12}$ F/m is the dielectric permittivity of vacuum, $c = 3 \times 10^8$ m/s is the speed of light, and P is the optical power carried along the waveguide by the unperturbed mode inside the waveguide. Considering the TE mode of the waveguide, we can assume that $\vec{\mathbf{E}}_{\text{opt}} \approx E_{\text{opt}} \hat{3}$. Furthermore, the dielectric perturbation tensor, $[\Delta \epsilon]$, is diagonal because the 1-2-3 coordinate

system is aligned with the coordinate system of the crystal and the DC E-field is applied in the extraordinary direction (see Appendix 1):

$$[\Delta\epsilon] = \begin{bmatrix} \Delta\epsilon_{11}^{(0)} & 0 & 0 \\ 0 & \Delta\epsilon_{22}^{(0)} & 0 \\ 0 & 0 & \Delta\epsilon_{33}^{(0)} \end{bmatrix}, \quad (3)$$

where

$$\begin{aligned} \Delta\epsilon_{11}^{(0)} &\approx -\epsilon_{11}^2(r_{13}E_3^{dc}) \\ \Delta\epsilon_{22}^{(0)} &\approx -\epsilon_{22}^2(r_{23}E_3^{dc}). \\ \Delta\epsilon_{33}^{(0)} &\approx -\epsilon_{33}^2(r_{33}E_3^{dc}) \end{aligned} \quad (4)$$

Therefore,

$$\Delta n_{\text{eff}}^{(0)} \approx \frac{c\epsilon_0}{4P} \Delta\epsilon_{33}^{(0)} \iint_{\text{core}} |E_{\text{opt}}|^2 dS. \quad (5)$$

The superscript (0) denotes the original state of zero rotation. Now we consider a case in which the waveguide is rotated by an angle ϕ as shown in Fig. 1. The change in the effective index of the optical mode in this case is denoted by $\Delta n_{\text{eff}}(\phi)$ and can be estimated according to the perturbation theory in the $1', 2', 3'$ coordinate system:

$$\Delta n_{\text{eff}}(\phi) \approx \frac{c\epsilon_0}{4P} \Delta\epsilon_{3'3'} \iint_{\text{core}} |E_{\text{opt}}|^2 dS \quad (6)$$

where we have assumed that the optical mode of the unperturbed waveguide is almost identical to the previous case with $\phi = 0$. Here, $\Delta\epsilon_{3'3'}$ is the (3,3) element of the dielectric perturbation tensor $[\Delta\epsilon]$ in the rotated $1', 2', 3'$ coordinate system. By comparing Eq. (5) with Eq. (6), we see that a relation between $\Delta n_{\text{eff}}(\phi)$ and $\Delta n_{\text{eff}}^{(0)}$ can be established if we find $\Delta\epsilon_{3'3'}$. To accomplish this, we consider the relation of the optical $\mathbf{E}_{\text{opt}}$ and $\mathbf{D}_{\text{opt}}$ fields:

$$\begin{bmatrix} D_{1'} \\ D_{3'} \end{bmatrix} = [\epsilon'] \begin{bmatrix} E_{1'} \\ E_{3'} \end{bmatrix}, \quad (7)$$

and the in-plane rotations

$$\begin{bmatrix} D_{1'} \\ D_{3'} \\ E_{1'} \\ E_{3'} \end{bmatrix} = \begin{bmatrix} \hat{1}' \cdot \hat{1} & \hat{1}' \cdot \hat{3} \\ \hat{3}' \cdot \hat{1} & \hat{3}' \cdot \hat{3} \\ \hat{1}' \cdot \hat{1} & \hat{1}' \cdot \hat{3} \\ \hat{3}' \cdot \hat{1} & \hat{3}' \cdot \hat{3} \end{bmatrix} \begin{bmatrix} D_1 \\ D_3 \\ E_1 \\ E_3 \end{bmatrix}. \quad (8)$$

Thus,

$$[\epsilon'] = R_\phi \begin{bmatrix} \epsilon_{11} & 0 \\ 0 & \epsilon_{33} \end{bmatrix} R_\phi^{-1}. \quad (9)$$

The electro-optic perturbation defined by $\Delta\mathbf{D}_{\text{opt}} = \Delta\epsilon \mathbf{E}_{\text{opt}}$ in the $1', 3'$ coordinate system is written as

$$[\Delta\epsilon'] = R_\phi \begin{bmatrix} \Delta\epsilon_{11} & 0 \\ 0 & \Delta\epsilon_{33} \end{bmatrix} R_\phi^{-1}, \quad (10)$$

where the rotation matrix is given by

$$R_\phi = \begin{bmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{bmatrix}. \quad (11)$$

Combining Eq. (10) and Eq. (11) results in

$$\Delta\epsilon_{3'3'} = \sin^2 \phi \Delta\epsilon_{11} + \cos^2 \phi \Delta\epsilon_{33}. \quad (12)$$

This key relation combined with Eq. (5) and Eq. (6) then yields

$$\Delta n_{\text{eff}}(\phi) \approx \Delta n_{\text{eff}}^{(0)} \left(\frac{\Delta\epsilon_{33}}{\Delta\epsilon_{33}^{(0)}} \cos^2 \phi + \frac{\Delta\epsilon_{11}}{\Delta\epsilon_{33}^{(0)}} \sin^2 \phi \right). \quad (13)$$

Noting that the applied DC field is $\mathbf{E}^{dc} = E^{dc} \hat{z}'$, and the ratio $\Delta\epsilon_{ii}/\Delta\epsilon_{33}^{(0)}$ can be simplified as

$$\begin{aligned} \left(\frac{\Delta\epsilon_{11}}{\Delta\epsilon_{33}^{(0)}} \right)_{Y\text{-cut}} &= \frac{\epsilon_{11}^2}{\epsilon_{33}^2} \left(\frac{r_{13}}{r_{33}} \cos \phi \right) \\ \left(\frac{\Delta\epsilon_{33}}{\Delta\epsilon_{33}^{(0)}} \right)_{Y\text{-cut}} &= \cos \phi \end{aligned} \quad (14)$$

because $r_{11} = 0$, $r_{31} = 0$, and $r_{32} = 0$. On the other hand, if we consider an X-cut crystal for which the in-plane axes are 2, 3 and rotated axes are 2', 3' then an equation similar to Eq. (13) still holds, but the ratios in Eq. (14) become

$$\begin{aligned} \left(\frac{\Delta\epsilon_{22}}{\Delta\epsilon_{33}^{(0)}} \right)_{X\text{-cut}} &= \frac{\epsilon_{22}^2}{\epsilon_{33}^2} \left(\frac{r_{22}}{r_{33}} \sin \phi + \frac{r_{23}}{r_{33}} \cos \phi \right) \\ \left(\frac{\Delta\epsilon_{33}}{\Delta\epsilon_{33}^{(0)}} \right)_{X\text{-cut}} &= \cos \phi \end{aligned} \quad (15)$$

because $r_{21} = 0$, $r_{31} = 0$, and $r_{32} = 0$. Finally, we note that due to the lack of large in-plane r_{ij} coefficients in the Z-cut crystal, modulators using lateral E-fields cannot produce performances comparable to their X and Y-cut counterparts. Thus, vertical E-fields are typically preferred for Z-cut modulators. For a Z-cut modulator with a vertical DC field inside the waveguide, the rotation has no effect on the electro-optic perturbation, i.e.,

$$\left(\frac{\Delta n_{\text{eff}}(\phi)}{\Delta n_{\text{eff}}^{(0)}} \right)_{Z\text{-cut}} = 1. \quad (16)$$

Table 1 summarizes the derived relations for the electro-optic perturbation under rotation for the LN crystals. It is worth noting that in this derivation the impact of rotation of the DC $[\epsilon]$ tensor on the applied DC field inside the LN waveguide has been ignored. A more accurate treatment that also includes the effect of rotation on the DC field is provided in Appendix 2.

Table 1. Electro-optic perturbation under rotation^a

Crystal	$\Delta n_{\text{eff}}(\phi)/\Delta n_{\text{eff}}^{(0)}$
X-cut	$\cos^3 \phi + \frac{\epsilon_{22}^2}{\epsilon_{33}^2} \left(\frac{r_{22}}{r_{33}} \sin \phi + \frac{r_{23}}{r_{33}} \cos \phi \right) \sin^2 \phi$
Y-cut	$\cos^3 \phi + \frac{\epsilon_{11}^2}{\epsilon_{33}^2} \left(\frac{r_{13}}{r_{33}} \cos \phi \right) \sin^2 \phi$
Z-cut	1

^a $\epsilon_{11} = \epsilon_{22} = 4.889 = \epsilon_o$ (ordinary, optical). $\epsilon_{33} = 4.569 = \epsilon_e$ (extraordinary, optical). $r_{13} = r_{23} = 8.6$ pm/V, $r_{22} = 3.4$ pm/V, $r_{33} = 30.8$ pm/V Z-cut assumes a vertical DC field.

It is worth pointing out that although the electro-optic performance of Z-cut waveguides does not suffer a reduction due to the rotation of the waveguide and the electrodes, X-cut and Y-cut

structures are easier to fabricate and show stronger electro-optic perturbation effects under the same applied voltages. This is due to the closer proximity of electrodes to the waveguide and higher optical confinement of the TE mode which result in the stronger interaction of optical TE mode of X-cut and Y-cut waveguides with the lateral DC electric field [4,10]. Typically, Z-cut waveguides require the TM mode to achieve high electro-optic perturbation, however, because of the low optical confinement of TM modes, the electrodes must be placed at a farther vertical distance above the waveguide [19], hence limiting the strength of the perturbation [20]. Hybrid approaches such as the one developed by Chen *et al.*, [21] can help mitigate this problem for Z-cut structures.

3. Electro-optic efficiency of microring modulators

In this section, we extend the analysis presented in the previous section to derive the electro-optic tuning efficiency of a microring resonator (or a race-track resonator) [10] designed in either X-cut or Y-cut crystals.

3.1. Tuning efficiency

Figure 2(a) shows the schematic of a microring resonator with one electrode inside and another electrode outside of the ring. The condition for the resonance of the microring resonator is that the round-trip phase shift inside the resonator is an integer multiple of 2π (i.e., $\Phi(\lambda) = 2m\pi$). Under the electro-optic perturbation, a particular resonance drifts away from its original value. However, the order of the resonance (i.e., m) remains the same. Hence, the variation of the round-trip phase shift is zero:

$$0 = \delta\Phi = \frac{d\Phi}{d\lambda} \times \Delta\lambda_{\text{res}} + \Delta\Phi_{EO}. \quad (17)$$

Using the definition of the phase shift

$$\Phi(\lambda) = \frac{2\pi}{\lambda} n_{\text{eff}} \times 2\pi R, \quad (18)$$

we see that [22]

$$\frac{d\Phi}{d\lambda} = -\frac{2\pi \times 2\pi R}{\lambda^2} n_g = -\frac{2\pi}{FSR_{nm}}, \quad (19)$$

where n_g is the optical group index of the unperturbed optical mode and FSR_{nm} is the free spectral range of the resonator (in the unit of nm). Combining Eq. (17) and Eq. (19) provides the shift of resonance:

$$\Delta\lambda_{\text{res}} = \frac{\Delta\Phi_{EO}}{2\pi} \times FSR_{nm}. \quad (20)$$

3.2. Filling factor

Due to the curvature of the microring resonator (Fig. 2(a)) and the electrodes, the applied DC field inside the ring deviates from the extraordinary direction as one moves along the ring from the angle $\phi = 0$. At a given angle ϕ , the induced electro-optic phase shift in a differential segment

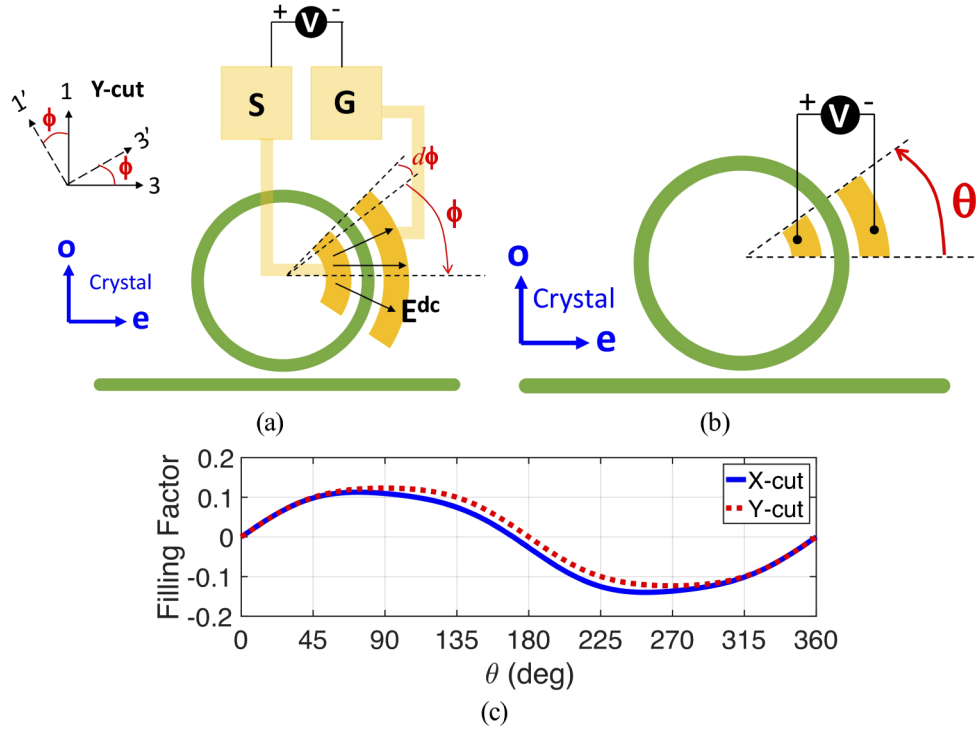


Fig. 2. (a) Schematic of a microring resonator in Y-cut LN. The curvature of the ring and electrodes causes the applied DC field inside the LN waveguide to deviate from the extraordinary direction ($\hat{3}$ axis). (b) Electrode arrangements for X-cut or Y-cut microring modulators with arbitrary extent originated from the extraordinary axis, and (c) its filling factor.

of the ring is given by:

$$\delta\Phi_{EO} = \frac{2\pi}{\lambda} \Delta n_{\text{eff}}(\phi) dl, \quad (21)$$

where $dl = R d\phi$. Therefore, the total induced electro-optic phase shift is calculated as

$$\Delta\Phi_{EO} = \frac{2\pi}{\lambda} R \int_{\phi_0}^{\phi_1} \Delta n_{\text{eff}}(\phi) d\phi, \quad (22)$$

which results in the following equation for the tuning efficiency

$$\frac{\Delta\lambda_{\text{res}}}{V_{DC}} = \left(\frac{\lambda_{\text{res}}}{n_g} \times \frac{\Delta n_{\text{eff}}^{(0)}}{V_{DC}} \right) \times ff. \quad (23)$$

Here, the term in parentheses is equivalent to a uniform electro-optic perturbation in the extraordinary direction (similar to a straight waveguide at the state of zero rotation) and the term ff given by

$$ff(\phi_0, \phi_1) = \frac{1}{2\pi} \int_{\phi_0}^{\phi_1} \frac{\Delta n_{\text{eff}}(\phi)}{\Delta n_{\text{eff}}^{(0)}} d\phi \quad (24)$$

represents the impact of the curvature of microring/electrodes and the filling extent of the electrodes (i.e., the fraction of the microring exposed to electro-optic perturbation). Using

Table 1, this filling factor can be explicitly calculated for X-cut and Y-cut microring resonators. Note that the filling factor for the straight waveguide phase-shifter in Fig. 1 is simply the factor in Table 1 because the waveguide has zero curvature. Based on Eq. (24), it is easy to show that the filling factor is maximum when $\phi_0 = -\pi/2$ and $\phi_1 = \pi/2$ for Y-cut. We can also see that the filling factor is exactly the same for X-cut and Y-cut microrings whenever $\phi_0 = -\phi_1$ because $r_{23} = r_{13}$. For X-cut, the maximum can be found numerically to occur when $\phi_0 = -73^\circ$ and $\phi_1 = 107^\circ$. The maximum possible value for the filling factor is 0.253 for X-cut and 0.246 for Y-cut. This establishes that the curvature of the microring resonators poses a fundamental limit on their electro-optic performance in X-cut or Y-cut thin-film LN platform. To further investigate the derived filling factor in Eq. (24) we consider three cases as shown in Figs. 2(b) and 3.

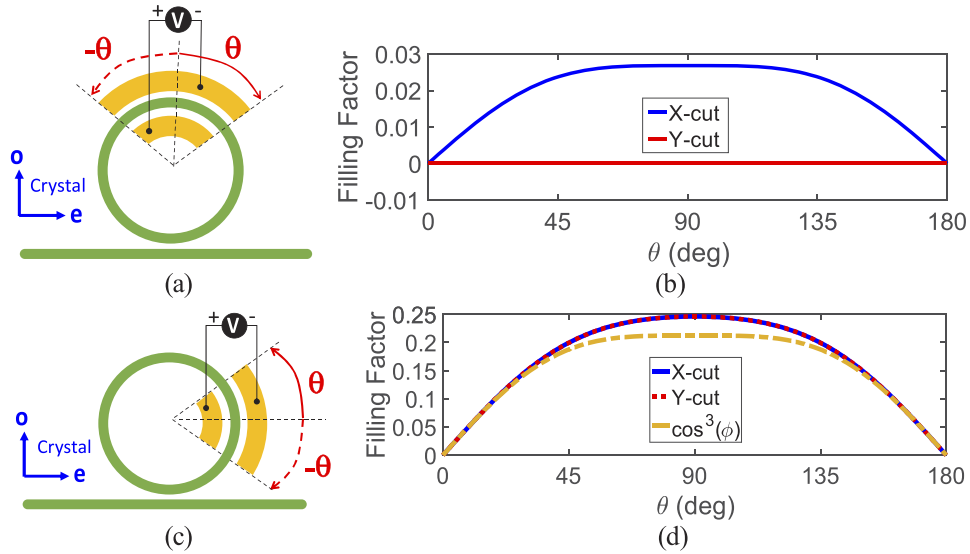


Fig. 3. (a) Symmetric extension of electrodes about the ordinary axis and (b) its filling factor. (c) Symmetric extension about the extraordinary axis, and (d) its filling factor.

Case 1: Shown in Fig. 2(b) where the electrodes are originated from the extraordinary direction and extended towards the ordinary direction of the crystal. Figure 2(c) shows the calculated filling factor as a function of the extension angle for both X-cut and Y-cut. As expected, a full extension ($\theta = 2\pi$) results in a *zero* filling factor, indicating that no net electro-optic phase shift can be accumulated inside the ring. It is also seen that the filling factors of X-cut and Y-cut LN crystals are slightly different which is attributed to the difference in their perturbation response under rotation as tabulated in Table 1.

Case 2: Shown in Fig. 3(a) where the electrodes are symmetrically extended about the in-plane ordinary axis of the crystal. The filling factors for X-cut and Y-cut crystals are calculated and plotted in Fig. 3(b) indicating that Y-cut design will yield no net electro-optic effect while the X-cut will yield a very weak effect with its maximum at $\theta = 90^\circ$.

Case 3: Shown in Fig. 3(c) where we symmetrically extend the electrodes about the extraordinary axis. The filling factors of X-cut and Y-cut crystals are plotted in Fig. 3(d) indicating the same results for both of them. The maximum performance is achieved at $\theta = 90^\circ$ with a filling factor of ~ 0.25 . Because this value is very close to the maximum possible value of the filling factor, we conclude that the electrode arrangement of Fig. 3(c) with $\theta \approx 90^\circ$ is the best possible design choice. We also plot the filling factor only due to the $\cos^3\phi$ term in Table 1 to show that the impact of r_{23} and r_{13} is in general not negligible.

Table 2 provides a summary of the demonstrated microring modulators based on thin-film lithium niobate. Although the X-cut and Y-cut modulators suffer from rotation effects, they consistently provide higher electro-optic tunability for the resonance of the microring. Z-cut structures only in the hybrid form [21] have been able to demonstrate good tunability.

Table 2. Literature review on the demonstrated tunability of X-cut, Y-cut, and Z-cut microring modulators

Author	Structure type	LN type	Mode	Tunability (pm/V)
Ahmed et al. [23]	hybrid LN/SiN	X-cut	TE	2.8
Zhang et al. [24]	partially etched LN	X-cut	TE	4
Rabiei et al. [16]	hybrid LN/Ta ₂ O ₅	X-cut	TE	4.5
Chen et al. [25]	hybrid LN/Si	X-cut	TE	3.3
Wang et al. [4]	partially etched LN	X-cut	TE	2.4
Wang et al. [10]	partially etched LN	X-cut	TE	7
Ahmed et al. [26]	hybrid LN/SiN	X-cut	TE	1.78
Rao et al. [9]	hybrid LN/Si ^a	Y-cut	TE	3.2
Mahmoud et al. [15]	partially etched LN	Y-cut	TE	0.32
Chen et al. [21]	hybrid LN/Si	Z-cut	TM	12.5
Wang et al. [27]	partially etched LN	Z-cut	TM	1.565
Guarino et al. [20]	partially etched LN	Z-cut	TM	1.05
Lee et al. [7]	hybrid LN/Si	Z-cut	TE	2
Siew et al. [28]	partially etched LN	Z-cut	TM	2.15
Krasnokutska et al. [19]	partially etched LN	Z-cut	TM	3

^awith chalcogenide glass

3.3. Segmented electrode design

Considering that the decrease in the filling factor in the electrode arrangement of Fig. 3(c) for $\theta > 90^\circ$ is due to the reversal of the horizontal component of the applied DC field, it is apparent that if the polarity of the DC field is reversed when $\theta > 90^\circ$, then the filling factor continues to increase to a maximum value of ~ 0.5 . This idea of segmented electrode design with a positive or negative polarity (while having the same voltage level) can be further generalized to the situation depicted in Fig. 4(a). Note that reversing the polarity of electrodes will cause the rotation factors in Table 1 to change sign in each differential segment shown in Fig. 2(a). Therefore, the goal is to find a continuous distribution of polarity (i.e., a function $f(\phi)$ with a value of +1 or -1) over the domain $-\pi \leq \phi \leq \pi$ such that the modified filling factor defined as

$$ff = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\Delta n_{\text{eff}}(\phi)}{\Delta n_{\text{eff}}^{(0)}} \times f(\phi) d\phi \quad (25)$$

is maximum. Considering the inequality

$$\int_{-\pi}^{\pi} G(x) dx \leq \int_{-\pi}^{\pi} |G(x)| dx \quad (26)$$

for any real-valued function, it is easy to conclude that the maximum of the filling factor occurs if and only if

$$f(\phi) = \text{sign} \left(\frac{\Delta n_{\text{eff}}(\phi)}{\Delta n_{\text{eff}}^{(0)}} \right) \quad (27)$$

which indicates that the optimal segmented electrode design depends on the number of zeros that the rotation factors in Table 1 have. Figure 4(b) shows the rotation factor of the X-cut crystal (blue curve) which has two zeros at $\phi = -73^\circ$ and $\phi = 107^\circ$.

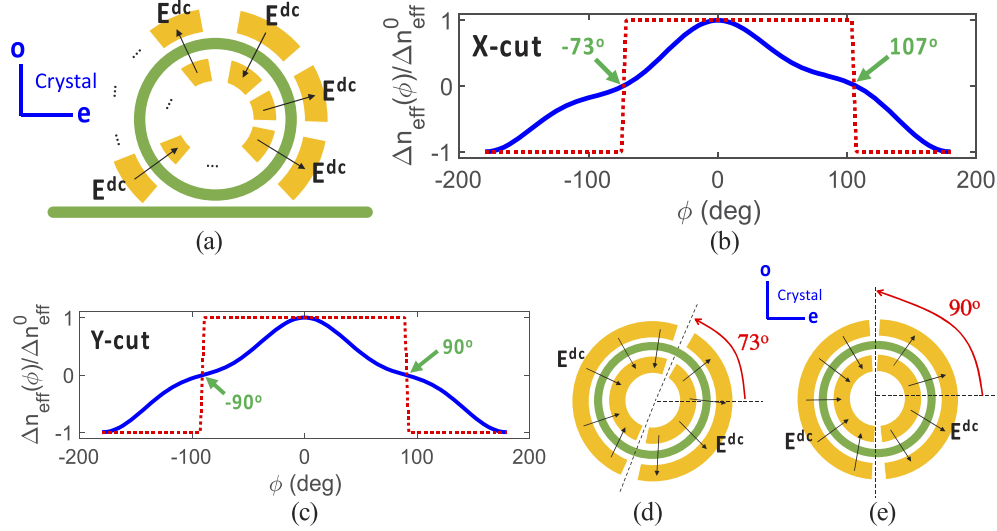


Fig. 4. (a) Schematic of a segmented electrode design for X-cut or Y-cut microring resonators. (b) Plot of X-cut rotation factor and its corresponding $f(\phi) = \pm 1$ function. (c) Plot of Y-cut rotation factor and its corresponding $f(\phi)$ function. (d) Optimal segmented electrode design for X-cut microrings. (e) Optimal segmented electrode design for Y-cut microrings.

The dotted red curve shows the function $f(\phi)$ with the values of ± 1 . Figure 4(c) shows the rotation factor and its corresponding $f(\phi) = \pm 1$ for the Y-cut crystal which is perfectly symmetric about $\phi = 0^\circ$. Therefore, the optimal segmented electrode design for X-cut microring resonators is the one shown in Fig. 4(d) which achieves a filling factor of 0.505. Similarly, the optimal segmented electrode design for Y-cut is the one shown in Fig. 4(e). It achieves a filling factor of 0.492. Both of these values are double the corresponding values for a single electrode.

3.4. Example 1: X-cut microring resonator

In order to further validate our analytic approach in Eqs. (23) and (24), we examine the structure of a microring resonator in X-cut demonstrated by Wang *et al.*, [10] with a measured electro-optic efficiency of ~ 7 pm/V. The cross-section of the partially etched thin-film LN is shown in Fig. 5(a) along with the dimensions used in the fabrication. Figure 5(b) shows the arrangement of electrodes used in [10]. Two pairs of symmetric electrodes (about the extraordinary direction) are used in this design. Because the polarity of the applied voltage is reversed for the electrodes on the left, each pair of electrodes contributes a filling factor of ~ 0.24 , and hence the overall filling factor is ~ 0.48 . Note that if the polarity of the electrodes were not reversed, one pair would contribute a filling factor of $+0.24$ whereas the other one would contribute -0.24 and the overall filling factor would be 0.

Figure 5(c) shows our finite element simulation (COMSOL software) of the applied DC electric field by applying 1V to the electrodes. The DC electric field inside the LN waveguide turns out to be horizontal which results in the strongest electro-optic interaction with the TE_{00} optical mode shown in Fig. 5(d). The effective index and group index of the optical mode are calculated and plotted in Fig. 5(e) and Fig. 5(f), respectively. At $\lambda = 1550$ nm, the effective index is ~ 1.927

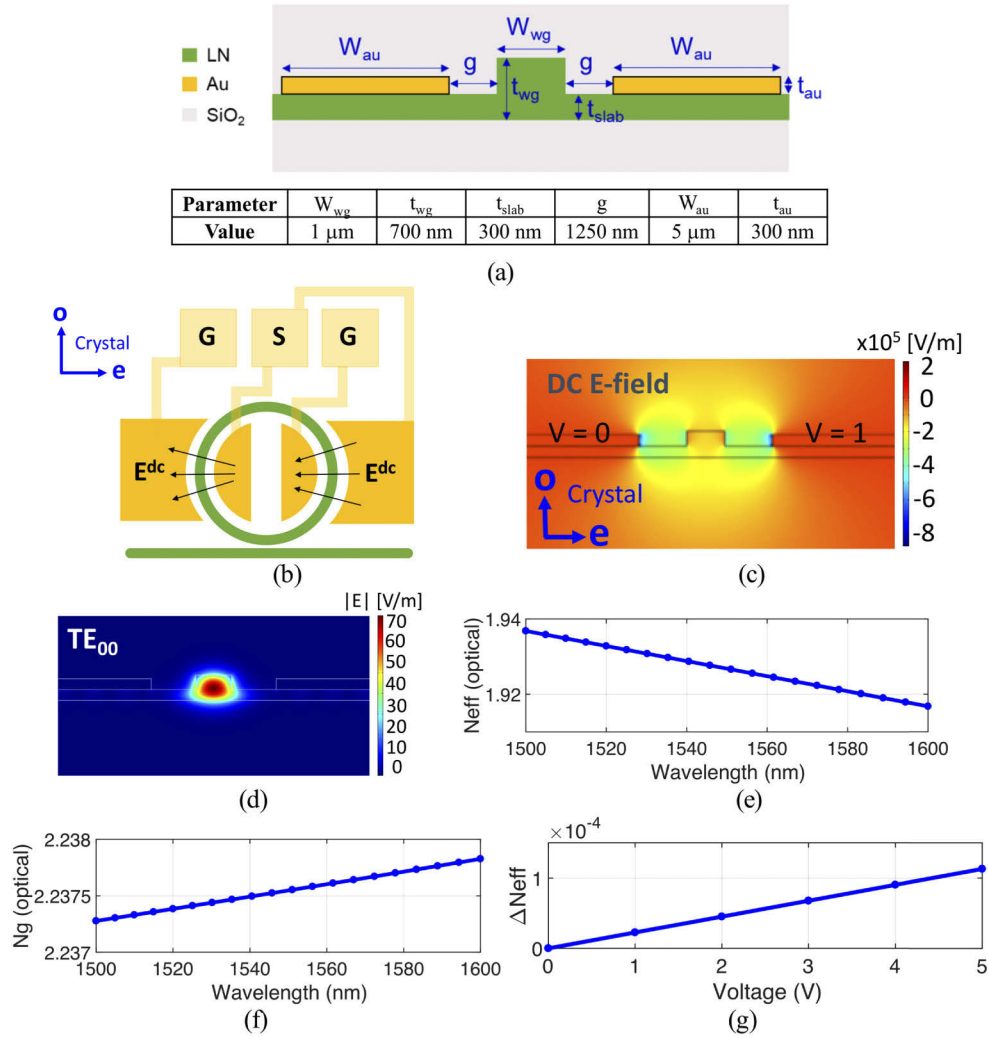


Fig. 5. (a) Cross-section of the LN waveguide used in [10] with its dimensions listed in the inset table. (b) Electrode arrangement used in the X-cut microring structure in [10]. The two pairs of electrodes have reverse polarity; hence each pair contributes a filling factor of ~ 0.24 and the overall filling factor is ~ 0.48 . (c) COMSOL simulation for the DC E-field distribution for $V_{\text{DC}} = 1$ V. The DC E-field inside the LN waveguide is almost horizontal. (d) COMSOL simulation of the fundamental TE_{00} optical mode of the waveguide (electric field). (e) Calculated optical effective index and (f) group index of the waveguide. (g) Change of effective index under different voltages.

and the group index is ~ 2.2375 . The electro-optic efficiency of the optical mode ($\Delta n_{\text{eff}}^{(0)}/V_{\text{DC}}$) is then calculated by sweeping the DC voltage. The results are plotted in Fig. 5(g). A linear behavior is clearly observed, yielding a modal electro-optic efficiency of

$$\frac{\Delta n_{\text{eff}}^{(0)}}{V_{\text{DC}}} \approx 2.258 \times 10^{-5} \text{ V}^{-1}, \quad (28)$$

and thus

$$\frac{\lambda_{\text{res}}}{n_g} \times \frac{\Delta n_{\text{eff}}^{(0)}}{V_{\text{DC}}} \approx 15.64 \text{ pm/V}. \quad (29)$$

Clearly, a value of 15.64 pm/V is far away from the actual reported measured value of 7 pm/V. However, if we include our analytic prediction of ~ 0.48 for the filling factor (curvature effect) in Eq. (23), the result is 7.5 pm/V which has a good agreement with the measurement. The slight disagreement is most likely due to the non-rectangular cross-section of the LN waveguide after fabrication which slightly alters the group index.

It is also worth mentioning that the electro-optic performance of the X-cut structure in Fig. 5(b) is sensitive to the rotation. For example, considering the result of Fig. 3(b), a 90° rotation will cause the filling factor of each pair of electrodes to change from 0.24 to 0.027, which is one order of magnitude reduction. This causes the tuning efficiency of the ring to vary from 7.5 pm/V to 0.75 pm/V. The same argument applies to Y-cut structures. One solution is to utilize Z-cut structures with an applied vertical field inside the LN waveguide [1,20,27,28]. This will remove all constraints on the orientation of the microring modulators.

3.5. Example 2: X-cut race-track resonator

The impact of the curvature on the electro-optic tuning efficiency of microrings can be reduced if straight sections are added to the resonator to create a race-track structure. We show in Appendix 3 that for a modulator with an arbitrary geometry that forms a closed path, the filling factor generalizes to

$$\mathcal{F} = \left(\frac{1}{\oint dl} \right) \oint \frac{\Delta n_{\text{eff}}(\phi)}{\Delta n_{\text{eff}}^{(0)}} dl. \quad (30)$$

This result can also be used to analyze the performance of Mach-Zehnder modulators under rotation [see Appendix 4]. Figure 6(a) shows the race-track structure with one pair of electrodes such that the straight parts are orthogonal to the extraordinary direction. The right straight part has a 180° rotation compared to the left straight part (see Fig. 1), hence setting $\phi = \pi$ in Table 1 results in the exact negative of the left straight side. The circular part has a filling factor of 0.027 according to Figs. 3(a) and 3(b) for the X-cut crystal. Therefore, the structure of Fig. 6(a) has a poor overall filling factor. The zero-sum contribution of the left and right straight parts can be negated if the polarity of the voltage applied to them is reversed. This is shown in Fig. 6(b). Equation (30) implies that the filling factor becomes

$$\mathcal{F} = \mathcal{F}_{\text{straight}} \frac{l_{\text{straight}}}{l_{\text{straight}} + l_{\text{circular}}} + \mathcal{F}_{\text{circular}} \frac{l_{\text{circular}}}{l_{\text{straight}} + l_{\text{circular}}}, \quad (31)$$

where l is the physical length of each segment. Because $\mathcal{F}_{\text{straight}} = 1$ and $\mathcal{F}_{\text{circular}} \leq 0.25$, the optimal configuration occurs with the electrode arrangement of Fig. 6(b) in the limit that the straight section is much longer than the curved section. The fundamental upper bound of the filling factor for race-track modulators is 1 and therefore it is impossible to go beyond Eq. (29) for the modulation efficiency in X-cut or Y-cut crystals.

As an example, we consider the race-track structure demonstrated by Wang *et al.*, [10] which has the same schematic as Fig. 6(b). The length of each straight segment is $\sim 280 \mu\text{m}$, and the

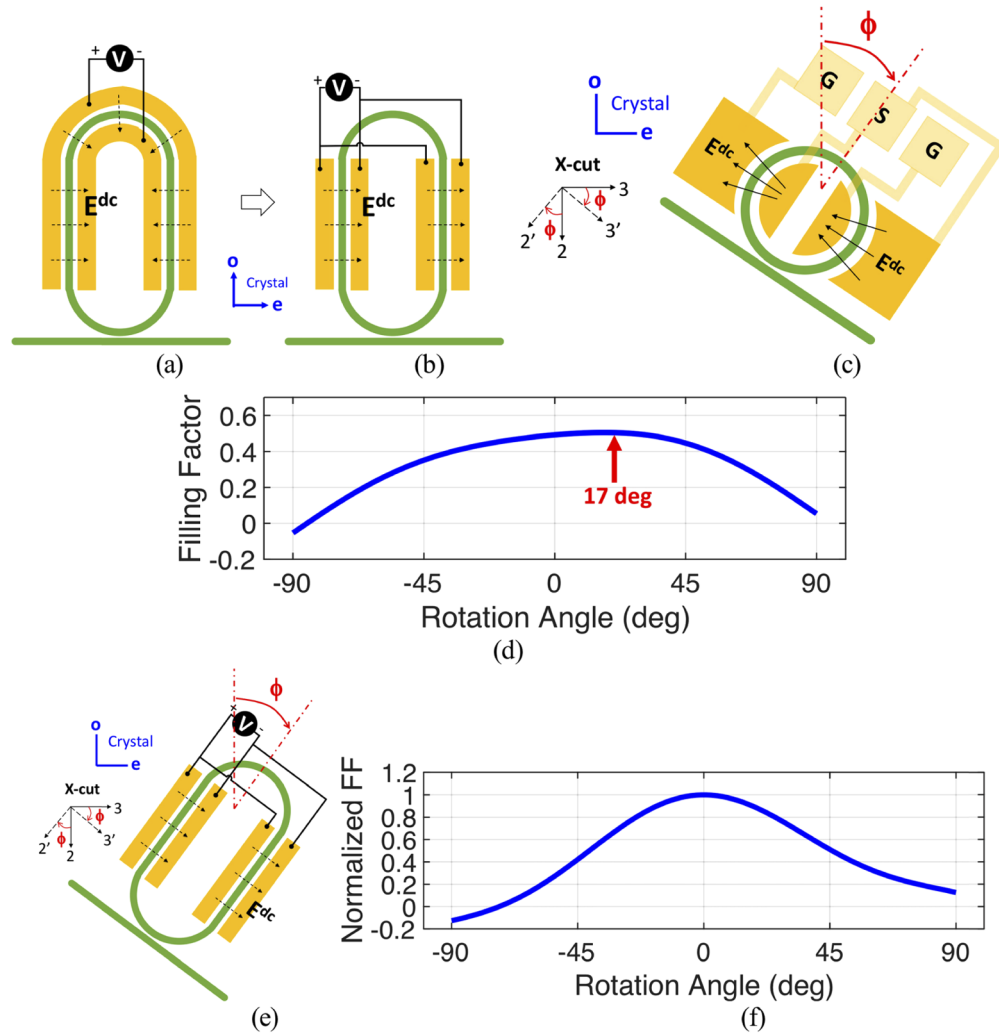


Fig. 6. (a) Schematic of a race-track resonator with a single pair of electrodes in X-cut or Y-cut. (b) Schematic of a race-track resonator with two pairs of electrodes. (c) X-cut microring modulator with double-pair electrodes under rotation. (d) Calculated filling factor under rotation showing an optimal angle of 17°. (e) Race-track modulator under rotation. (f) Normalized filling factor for the race-track resonator under rotation.

length of each half circular part is $\sim 280 \mu\text{m}$; hence the filling factor is ~ 0.5 which combined with Eq. (29) results in an electro-optic tuning efficiency of $\sim 7.8 \text{ pm/V}$. This analytic result agrees well with the reported measured value of $\sim 7 \text{ pm/V}$. Table 3 summarizes the comparison of our analytic results with the reported measured values in [10].

Table 3. Comparison of measured and theoretical estimates of electro-optic efficiency

Structure	Crystal	Measured	Our analysis
Microring	X-cut	7 pm/V [10]	7.48 pm/V
Race-track	X-cut	7 pm/V [10]	7.82 pm/V

3.6. X-cut microring and race-track under rotation

The term proportional to $r_{22}\sin^3\phi$ in Table 1 for the X-cut electro-optic efficiency can be positive or negative depending on the rotation. Figure 6(c) shows the microring resonator with double-pair electrode that has been rotated by an angle ϕ . Figure 6(d) shows the calculated filling factor as a function of the rotation angle. The maximum value of the filling factor is slightly larger than the maximum value in Fig. 6(d) and happens at a rotation angle of 17° . Figure 6(e) shows the structure of the race-track modulator under the rotation and Fig. 6(f) shows the value of filling factor normalized to its value from Eq. (31) when $\phi = 0$. As expected, a rotation angle of 0° gives the maximum value in the race-track case.

4. Conclusions

Based on the perturbation theory of optical waveguides, we rigorously derive the relations for the change of effective index under the rotation of the electrodes for both X-cut and Y-cut structures in thin-film LN platform. We then show that the tuning efficiency of microring resonators in X-cut and Y-cut crystals has a fundamental upper bound due to the curvature of electrodes which is quantified in terms of a coefficient called the filling factor (maximum ~ 0.25 for a single electrode pair). We also show that a microring modulator with two pairs of electrodes in opposite polarities can achieve a maximum filling factor of ~ 0.5 . Solutions to these limitations are to either use Z-cut microring modulators with a vertical DC field or X-cut or Y-cut race-track modulators with segmented electrodes. Comparison of our analytical results with the measured results in the literature demonstrates the accuracy of our approach.

Appendix 1: derivation of $[\Delta\epsilon]$ tensor

In order to find the electro-optic perturbation of LN, we start by writing the relation between the optical E-field and dielectric polarization as

$$\mathbf{E}_{opt} = [\epsilon]^{-1}\mathbf{D}_{opt} = [\eta]\mathbf{D}_{opt} = \begin{bmatrix} \eta_{11} & \eta_{12} & \eta_{13} \\ \eta_{21} & \eta_{22} & \eta_{23} \\ \eta_{31} & \eta_{32} & \eta_{33} \end{bmatrix} \mathbf{D}_{opt}, \quad (32)$$

where $[\epsilon]$ and $[\eta]$ are the optical permittivity and impermeability tensors, respectively. The perturbation of the impermeability tensor, $[\Delta\eta] = [\eta] - [\eta_0]$, is a 3×3 matrix given by the standard definition of the electro-optic coefficients r_{ij} of LN material [29]:

$$\Delta\eta = \begin{bmatrix} r_{12}E_2^{DC} + r_{13}E_3^{DC} & r_{61}E_1^{DC} & r_{51}E_1^{DC} \\ r_{61}E_1^{DC} & r_{22}E_2^{DC} + r_{23}E_3^{DC} & r_{42}E_2^{DC} \\ r_{51}E_1^{DC} & r_{42}E_2^{DC} & r_{33}E_3^{DC} \end{bmatrix}. \quad (33)$$

Next, we consider the optical permittivity tensor of the LN, $[\epsilon]$, including the perturbation due to the presence of the electrostatic field. Let $[\epsilon_0]$ be the unperturbed optical permittivity tensor of LN and I be the identity matrix. To find $[\epsilon]$, we can write

$$\begin{aligned} [\epsilon] &= [\epsilon_0] + [\Delta\epsilon] = ([\eta_0] + [\Delta\eta])^{-1} \\ &= (I + [\eta_0]^{-1} \cdot [\Delta\eta])^{-1} \cdot [\eta_0]^{-1} = (I + [\epsilon_0] \cdot [\Delta\eta])^{-1} \cdot [\epsilon_0], \end{aligned} \quad (34)$$

where we used $(AB)^{-1} = B^{-1}A^{-1}$ for invertible matrices $B = I + [\eta_0]^{-1} \cdot [\Delta\eta]$ and $A = [\eta_0]$. Therefore:

$$[\Delta\epsilon] = ((I + [\epsilon_0] \cdot [\Delta\eta])^{-1} - I) \cdot [\epsilon_0]. \quad (35)$$

To the first-order approximation, we can write

$$(I + [\epsilon_0] \cdot [\Delta\eta])^{-1} \approx I - [\epsilon_0] \cdot [\Delta\eta], \quad (36)$$

hence:

$$[\Delta\epsilon] \approx -[\epsilon_0] \cdot [\Delta\eta] \cdot [\epsilon_0]. \quad (37)$$

The ij component of the $[\Delta\epsilon]$ tensor is then calculated as

$$\Delta\epsilon_{ij} = - \sum_k \sum_l [\epsilon_0]_{ik} [\Delta\eta]_{kl} [\epsilon_0]_{lj}, \quad (38)$$

where the summation is on k and l indices. Due to the diagonal form of the $[\epsilon_0]$ tensor, $[\epsilon_0]_{ij} = \epsilon_{ii} \delta_{ij}$ where δ_{ij} is the Kronecker delta. Therefore, $[\Delta\epsilon]$ tensor has the following form:

$$[\Delta\epsilon] \approx - \begin{bmatrix} \epsilon_{11}^2 \Delta\eta_{11} & \epsilon_{11} \epsilon_{22} \Delta\eta_{12} & \epsilon_{11} \epsilon_{33} \Delta\eta_{13} \\ \epsilon_{11} \epsilon_{22} \Delta\eta_{21} & \epsilon_{22}^2 \Delta\eta_{22} & \epsilon_{22} \epsilon_{33} \Delta\eta_{23} \\ \epsilon_{11} \epsilon_{33} \Delta\eta_{31} & \epsilon_{22} \epsilon_{33} \Delta\eta_{32} & \epsilon_{33}^2 \Delta\eta_{33} \end{bmatrix}, \quad (39)$$

where $\epsilon_{11} = \epsilon_{22} = \epsilon_{\text{ordinary}}$ and $\epsilon_{33} = \epsilon_{\text{extraordinary}}$. If the DC E-field is applied in the extraordinary direction as in Fig. 1, then $E_1^{\text{dc}} = 0$, $E_2^{\text{dc}} = 0$, and $E_3^{\text{dc}} = E^{\text{dc}}$; therefore $\Delta\eta_{12} = \Delta\eta_{21} = \Delta\eta_{13} = \Delta\eta_{31} = \Delta\eta_{23} = \Delta\eta_{32} = 0$ and $[\Delta\epsilon]$ remains a diagonal matrix as given by Eq. (3).

Appendix 2: finite element simulation of rotation factors

The impact of rotation on the applied DC field inside the LN waveguide in Fig. 5(a) can be approximated using a capacitive circuit model presented in Fig. 7(a). Considering that the DC field mostly resides outside the LN and is almost horizontal, the capacitors in the cross-section (per unit length) are defined only for the highlighted red region in Fig. 7(a) such that:

$$C_{\text{SiO}_2} \approx \epsilon_{\text{SiO}_2} \frac{t_{\text{wg}} - t_{\text{slab}}}{g}, \quad C_{\text{slab}} \approx \epsilon_{3'3'} \frac{t_{\text{slab}}}{g}, \quad C_{\text{wg}} \approx \epsilon_{3'3'} \frac{t_{\text{wg}}}{w_{\text{wg}}}, \quad (40)$$

where $\epsilon_{\text{SiO}_2} = 3.9$ and $\epsilon_{3'3'}$ is the rotated version of ϵ_{33} (DC value). According to Eq. (9) the value of $\epsilon_{3'3'}$ is calculated as

$$\epsilon_{3'3'} = \epsilon_{33} \cos^2 \phi + \epsilon_{22} \sin^2 \phi \quad (41)$$

for the X-cut and

$$\epsilon_{3'3'} = \epsilon_{33} \cos^2 \phi + \epsilon_{11} \sin^2 \phi \quad (42)$$

for the Y-cut with $\epsilon_{11} = \epsilon_{22} = 46.5$ and $\epsilon_{33} = 27.3$. The DC field inside of the LN waveguide can then be written as

$$E^{\text{dc}}(\phi) = \frac{V_{\text{dc}}}{w_{\text{wg}}} \times \frac{1}{1 + 2C_{\text{wg}}/(C_{\text{SiO}_2} + C_{\text{slab}})}. \quad (43)$$

Figure 7(b) shows the calculated change in the strength of DC field inside the LN waveguide as a function of rotation angle (solid curve) along with the finite element simulation result in COMSOL (dotted curve) for $V_{\text{dc}} = 1$ V. The parameters are set according to Fig. 5(a). To find the applied DC field inside the LN waveguide under rotation in COMSOL, we setup an electrostatic simulation with the rotated DC $[\epsilon]$ tensor of LN according to Eq. (9) for either X-cut or Y-cut

crystals. For example, for the X-cut the rotated $[\epsilon']$ tensor has a form of

$$[\epsilon'] = \begin{bmatrix} \epsilon_{11} & 0 & 0 \\ 0 & \epsilon_{2'2'} & \epsilon_{2'3'} \\ 0 & \epsilon_{3'2'} & \epsilon_{3'3'} \end{bmatrix} \quad (44)$$

where

$$\begin{aligned} \epsilon_{2'2'} &= \epsilon_{22} \cos^2 \phi + \epsilon_{33} \sin^2 \phi \\ \epsilon_{2'3'} &= \epsilon_{3'2'} = (\epsilon_{33} - \epsilon_{22}) \sin \phi \cos \phi. \end{aligned} \quad (45)$$

As shown in Fig. 7(b), the result of capacitor model is a good approximation of the results of COMSOL, both of which confirming that the rotation does not impose a significant change on the applied DC field inside the LN waveguide.

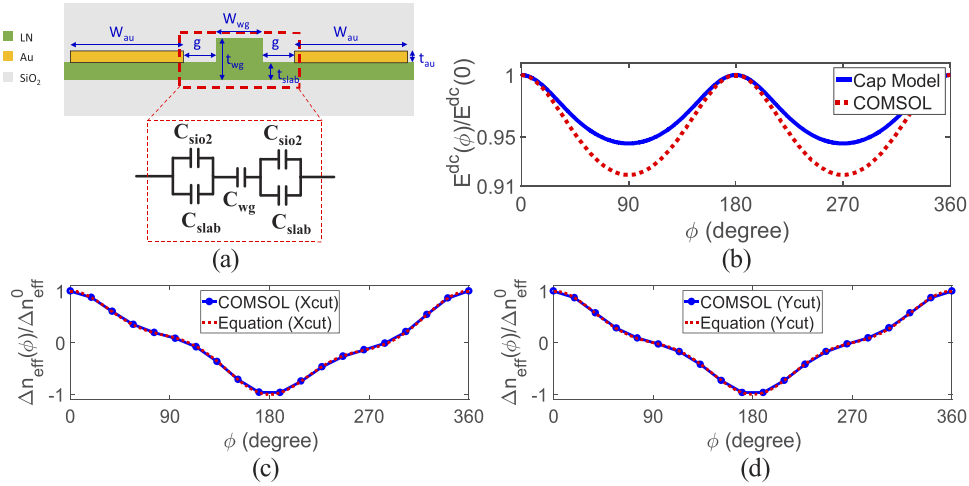


Fig. 7. (a) Cross-section of the LN waveguide and electrodes at a rotated plane (ϕ) and its approximate capacitor equivalent circuit for the distribution of the DC field. (b) Comparison of the estimated rotated DC field inside the LN waveguide with COMSOL. (c) Comparison of the X-cut rotation factors calculated from Table 1 with the correction on DC field (dotted) and directly from COMSOL (solid). The zero-crossing points are $\phi = 107^\circ$ and $\phi = -73^\circ$. (d) Comparison of the Y-cut rotation factors calculated from Table 1 with the correction on DC field (dotted) and directly from COMSOL (solid). The zero-crossing points are $\phi = 90^\circ$ and $\phi = -90^\circ$.

We next proceed to calculate the electro-optic perturbation under rotation by multiplying the estimated ratio of $E^{dc}(\phi)/E^{dc}(0)$ from Eq. (43) with the rotation factors in Table 1 to correct for the change in the DC field. Considering that the minima of $E^{dc}(\phi)/E^{dc}(0)$ occur at angles for which the rotation factors in Table 1 are almost zero, it is reasonable to expect that the final results should be very similar to the plots of Figs. 4(b) and 4(c). The plots of corrected rotation factors along with the results of finite element analysis shown in Figs. 7(c) and 7(d) confirm this expectation. In order to find the modal electro-optic perturbation under rotation in COMSOL the following steps were taken:

- **Step 1:** rotate the optical $[\epsilon]$ tensor in the absence of applied DC field ($V_{dc} = 0$ V) and find the optical mode in the cross-section (2D simulation). The results are $n_{eff}^0(\phi)$.

- **Step 2:** rotate the DC $[\epsilon]$ tensor of LN and calculate the electro-static field distribution in the cross-section. The results are $\mathbf{E}^{dc} = E_{3'}\hat{3}' + E_{1'}\hat{1}'$ for the X-cut and $\mathbf{E}^{dc} = E_{3'}\hat{3}' + E_{2'}\hat{2}'$ for the Y-cut.
- **Step 3:** perturb the optical $[\epsilon]$ tensor of LN in the crystal coordinate system using Eq. (3) by expressing the calculated DC field in the crystal coordinate system.
- **Step 4:** rotate the perturbed optical $[\epsilon]$ and calculate the optical mode. The results are $n_{\text{eff}}(\phi)$.
- **Step 5:** define $\Delta n_{\text{eff}}(\phi) = n_{\text{eff}}(\phi) - n_{\text{eff}}^0(\phi)$ and normalize it to its value at $\phi = 0$ to get the rotation factors in Figs. 7(c) and 7(d).

The comparison of the rotation factors from Table 1 including the correction on DC field (Eq. (43)) with the results of COMSOL shows a good match; hence, the accuracy of our analytical approach is good.

Appendix 3: general definition of the filling factor

To derive the filling factor for electrodes surrounding an arbitrary closed-path LN waveguide, we start by writing the total phase accumulated traversing the path as the scalar sum

$$\Phi = \oint \frac{2\pi}{\lambda} n_{\text{eff}} dl \quad (46)$$

where dl is the scalar length of the differential segment. Applying Eq. (46) to Eq. (17), we generalize the result of Eq. (20):

$$\Delta\lambda_{\text{res}} = \frac{\lambda^2}{2\pi} \frac{1}{\oint \left(n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda} \right) dl} \Delta\Phi_{EO}. \quad (47)$$

The incremental phase shift in a differential segment of the waveguide due to the electro-optic effect is

$$\delta\Phi_{EO} = \frac{2\pi}{\lambda} \Delta n_{\text{eff}}(\phi) dl, \quad (48)$$

and thus, the total electro-optic phase shift for the path is

$$\Delta\Phi_{EO} = \frac{2\pi}{\lambda} \oint \Delta n_{\text{eff}}(\phi) dl. \quad (49)$$

Defining $n_g = \oint (n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}) dl / \oint dl$ as the average group index along the path and inserting Eq. (49) into Eq. (47) yields

$$\Delta\lambda_{\text{res}} = \frac{\lambda}{n_g} \frac{1}{\oint dl} \oint \Delta n_{\text{eff}}(\phi) dl. \quad (50)$$

We next multiply and divide this result by the constant $\Delta n_{\text{eff}}^{(0)}$, which refers to the electro-optic effect of a waveguide with electrodes oriented normal to the extraordinary direction so that the DC electric field and the optical mode interact with each other based on the r_{33} coefficient. Hence, our final result for the tuning efficiency of the arbitrary closed path resonator is

$$\frac{\Delta\lambda_{\text{res}}}{V^{DC}} = \left(\frac{\lambda_{\text{res}}}{n_g} \times \frac{\Delta n_{\text{eff}}^{(0)}}{V^{DC}} \right) \left(\frac{1}{\oint dl} \oint \frac{\Delta n_{\text{eff}}(\phi)}{\Delta n_{\text{eff}}^{(0)}} dl \right). \quad (51)$$

The first factor is equivalent to the tuning efficiency of a resonator whose electro-optic effect is in the extraordinary direction at each point along the path. The second term is the filling factor that

describes the curvature effect:

$$ff = \frac{1}{\oint dl} \oint \frac{\Delta n_{\text{eff}}(\phi)}{\Delta n_{\text{eff}}^{(0)}} dl. \quad (52)$$

Each integral here can be divided into integrals along straight sections of the path that can easily be computed because the integrand is constant plus integrals along curved sections that can be computed using the definition of the curvature $C = |d\theta/dl|$ [30] where θ is the angle of curvature with respect to the horizontal axis, i.e., $\theta = \pi/2 - \phi$; see Fig. 8(a). Thus, for curved sections, $dl = d\phi/C(\phi)$. Because a fully circular microring resonator has a constant curvature, i.e., $C(\phi) = 1/R$, Eq. (52) reduces to Eq. (24) as expected.

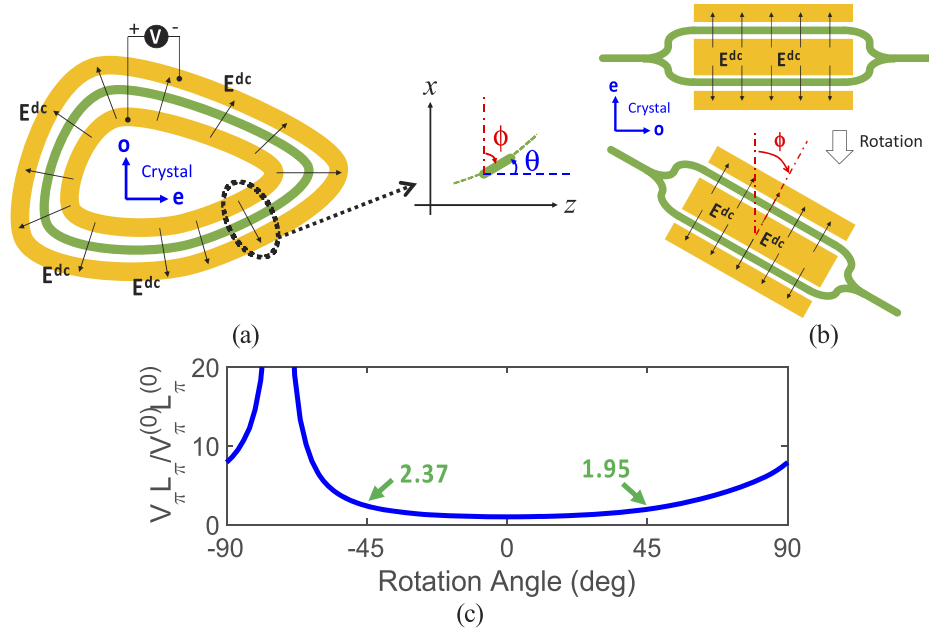


Fig. 8. (a) Arbitrary closed path LN waveguide with electrodes. The differential segment of length, dl , is shown on the right. (b) Structure of a push-pull Mach-Zehnder modulator and its rotated version. (c) Normalized value of $V_{\pi} L_{\pi}$ under rotation for the X-cut LN crystal.

Appendix 4: Mach-Zehnder modulator under rotation

The rotation factors in Table 1 can also be used to investigate the electro-optical behavior of Mach-Zehnder modulators under rotation (e.g., in a push-pull design as shown in Fig. 8(b)). For a given arm of the Mach-Zehnder (e.g., top arm in Fig. 8(b)) of which the applied DC E-field is rotated by angle ϕ with respect to the extraordinary direction of the LN crystal, the accumulated phase shift is

$$\Delta\Phi_{EO}^{+} = \frac{2\pi}{\lambda} \Delta n_{\text{eff}}^{(0)} \int \frac{\Delta n_{\text{eff}}(\phi)}{\Delta n_{\text{eff}}^{(0)}} dl, \quad (53)$$

and because the other arm is a 180° rotated version of this arm, its phase shift is

$$\Delta\Phi_{EO}^{-} = \frac{2\pi}{\lambda} \Delta n_{\text{eff}}^{(0)} \int \frac{\Delta n_{\text{eff}}(\pi + \phi)}{\Delta n_{\text{eff}}^{(0)}} dl. \quad (54)$$

In general, $\Delta n_{\text{eff}}(\pi + \phi) = -\Delta n_{\text{eff}}(\phi)$ for both X-cut and Y-cut. Assuming the electrodes are only on the straight sections of the modulator, the total phase shift difference is then given by

$$\Delta\Phi = \Delta\Phi_{\text{max}} \times \frac{\Delta n_{\text{eff}}(\phi)}{\Delta n_{\text{eff}}^{(0)}}, \quad (55)$$

where $\Delta\Phi_{\text{max}} = 4\pi l \Delta n_{\text{eff}}^{(0)} / \lambda$. Finally, using these equations, the $V_{\pi}L_{\pi}$ metric of the rotated Mach-Zehnder modulator is

$$V_{\pi}(\phi) \cdot L_{\pi}(\phi) = \frac{V_{\pi}^{(0)} \cdot L_{\pi}^{(0)}}{\Delta n_{\text{eff}}(\phi) / \Delta n_{\text{eff}}^{(0)}}, \quad (56)$$

which indicates that rotation increases the $V_{\pi}L_{\pi}$ metric and degrades the performance. Figure 8(c) shows the plot of $V_{\pi}(\phi)L_{\pi}(\phi)$ normalized to its maximum value at $\phi = 0$ for the X-cut. We see that a 45° rotation increases $V_{\pi}L_{\pi}$ (DC value) by a factor of 1.95, whereas a rotation of -45° results in an increase of 2.37. The difference is due to the $r_{22}\sin^3\phi$ term. Furthermore, we see that at $\phi = -73^{\circ}$, the normalized value of $V_{\pi}L_{\pi}$ goes to infinity due to $\Delta n_{\text{eff}}(\phi) = 0$.

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References

1. M. Bahadori, A. Kar, Y. Yang, A. Lavasani, L. Goddard, and S. Gong, "High-performance integrated photonics in thin film lithium niobate platform," in *Conference on Lasers and Electro-Optics* (2019).
2. A. Kar, M. Bahadori, S. Gong, and L. L. Goddard, "Realization of alignment-tolerant grating couplers for z-cut thin-film lithium niobate," *Opt. Express* **27**(11), 15856–15867 (2019).
3. A. Rao and S. Fathpour, "Second-harmonic generation in integrated photonics on silicon," *Phys. Status Solidi A* **215**(4), 1700684 (2018).
4. C. Wang, M. Zhang, M. Yu, R. Zhu, H. Hu, and M. Loncar, "Monolithic lithium niobate photonic circuits for Kerr frequency comb generation and modulation," *Nat. Commun.* **10**(1), 978 (2019).
5. I. of E. Engineers, *Properties of Lithium Niobate* (IET, 2002).
6. A. Boes, B. Corcoran, L. Chang, J. Bowers, and A. Mitchell, "Status and potential of lithium niobate on insulator (LNOI) for photonic integrated circuits," *Laser Photonics Rev.* **12**(4), 1700256 (2018).
7. Y. S. Lee, G.-D. Kim, W.-J. Kim, S.-S. Lee, W.-G. Lee, and W. H. Steier, "Hybrid Si-LiNbO₃ microring electro-optically tunable resonators for active photonic devices," *Opt. Lett.* **36**(7), 1119–1121 (2011).
8. A. Rao, A. Patil, P. Rabiei, A. Honardoost, R. DeSalvo, A. Paolella, and S. Fathpour, "High-performance and linear thin-film lithium niobate Mach-Zehnder modulators on silicon up to 50 GHz," *Opt. Lett.* **41**(24), 5700–5703 (2016).
9. A. Rao, A. Patil, J. Chiles, M. Malinowski, S. Novak, K. Richardson, P. Rabiei, and S. Fathpour, "Heterogeneous microring and Mach-Zehnder modulators based on lithium niobate and chalcogenide glasses on silicon," *Opt. Express* **23**(17), 22746–22752 (2015).
10. C. Wang, M. Zhang, B. Stern, M. Lipson, and M. Loncar, "Nanophotonic lithium niobate electro-optic modulators," *Opt. Express* **26**(2), 1547–1555 (2018).
11. M. Prost, G. Liu, and S. J. B. Yoo, "A Compact Thin-Film Lithium Niobate Platform with Arrayed Waveguide Gratings and MMIs," in *2018 Optical Fiber Communications Conference and Exposition (OFC)* (2018), pp. 1–3.
12. C. Wang, M. Zhang, X. Chen, M. Bertrand, A. Shams-Ansari, S. Chandrasekhar, P. Winzer, and M. Loncar, "Integrated lithium niobate electro-optic modulators operating at CMOS-compatible voltages," *Nature* **562**(7725), 101–104 (2018).
13. B. Desiatov, A. Shams-Ansari, M. Zhang, C. Wang, and M. Loncar, "Ultra-low-loss integrated visible photonics using thin-film lithium niobate," *Optica* **6**(3), 380–384 (2019).
14. A. J. Mercante, S. Shi, P. Yao, L. Xie, R. M. Weikle, and D. W. Prather, "Thin film lithium niobate electro-optic modulator with terahertz operating bandwidth," *Opt. Express* **26**(11), 14810–14816 (2018).
15. M. Mahmoud, L. Cai, C. Bottenfield, and G. Piazza, "Lithium niobate electro-optic racetrack modulator etched in Y-cut LNOI platform," *IEEE Photonics J.* **10**(1), 1–10 (2018).
16. P. Rabiei, J. Ma, S. Khan, J. Chiles, and S. Fathpour, "Heterogeneous lithium niobate photonics on silicon substrates," *Opt. Express* **21**(21), 25573–25581 (2013).

17. A. Rao and S. Fathpour, "Compact Lithium Niobate Electrooptic Modulators," *IEEE J. Sel. Top. Quantum Electron.* **24**(4), 3400114 (2018).
18. C. R. Pollock and M. Lipson, *Integrated Photonics* (Springer, 2003).
19. I. Krasnokutska, J.-L. J. Tambasco, and A. Peruzzo, "Tunable large free spectral range microring resonators in lithium niobate on insulator," *Sci. Rep.* **9**(11086), (2019).
20. A. Guarino, G. Poberaj, D. Rezzonico, R. Degl'Innocenti, and P. Günter, "Electro-optically tunable microring resonators in lithium niobate," *Nat. Photonics* **1**(7), 407–410 (2007).
21. L. Chen, M. G. Wood, and R. M. Reano, "12.5 pm/V hybrid silicon and lithium niobate optical microring resonator with integrated electrodes," *Opt. Express* **21**(22), 27003 (2013).
22. M. Bahadori, M. Nikdast, S. Rumley, L. Y. Dai, N. Janosik, T. V. Vaerenbergh, A. Gazman, Q. Cheng, R. Polster, and K. Bergman, "Design space exploration of microring resonators in silicon photonic interconnects: impact of the ring curvature," *J. Lightwave Technol.* **36**(13), 2767–2782 (2018).
23. A. N. R. Ahmed, S. Shi, J. Manley, P. Yao, and D. W. Prather, "Electro-Optically Tunable Modified Racetrack Resonator in Hybrid Si₃N₄-LiNbO₃," *Adv. Photonics Congr.*, **2** (2019).
24. M. Zhang, C. Wang, Y. Hu, A. Shams-Ansari, T. Ren, S. Fan, and M. Lončar, "Electronically programmable photonic molecule," *Nat. Photonics* **13**(1), 36–40 (2019).
25. L. Chen, Q. Xu, M. G. Wood, and R. M. Reano, "Hybrid silicon and lithium niobate electro-optical ring modulator," *Optica* **1**(2), 112–118 (2014).
26. A. N. R. Ahmed, S. Shi, M. Zablocki, P. Yao, and D. W. Prather, "Tunable hybrid silicon nitride and thin-film lithium niobate electro-optic microresonator," *Opt. Lett.* **44**(3), 618–621 (2019).
27. T.-J. Wang, C.-H. Chu, and C.-Y. Lin, "Electro-optically tunable microring resonators on lithium niobate," *Opt. Lett.* **32**(19), 2777–2779 (2007).
28. S. Y. Siew, S. S. Saha, M. Tsang, and A. J. Danner, "Rib Microring Resonators in Lithium Niobate on Insulator," *IEEE Photonics Technol. Lett.* **28**(5), 573–576 (2016).
29. B. E. A. Saleh and M. C. Teich, *Fundamentals of Photonics* (John Wiley & Sons, 2019).
30. M. Bahadori, M. Nikdast, Q. Cheng, and K. Bergman, "Universal Design of Waveguide Bends in Silicon-on-Insulator Photonics Platform," *J. Lightwave Technol.* **37**(13), 3044–3054 (2019).